CS-003 Mold Design & Manufacturing

Doc ID	CS-003
Title	Mold Design & Manufacturing (Tooling Board)
Revision	D
Status	Draft, pending Lead signature
Effective date	2026-08-28
Supersedes	Tribal mold-design practice; consolidates *SN4 Mold Manufacturing Problems and Solutions* + *Mold Design Documentation*

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial issue
B	2026-07-28	Claude	Language pass: fewer em dashes for rhythm variety, no content change.
C	2026-08-03	Claude	Glue-up marking made explicit in §7.3: oil-based paint marker on two adjacent side faces, not a Sharpie, which the process's own IPA wipe dissolves (CS-001 §7.4).
D	2026-08-28	Claude	Engineering pass (CS-000 Rev C conventions). Figure 1 dimensions the §7.1 design rules on a mold cross-section, which is the form the §7.2 reviewer actually checks them in. No rule changed.

1. Purpose

A mold error you find after machining burns our three scarcest things at once: tooling board, machine time, and calendar. The SN5 clamshell CAD error (raised middle section, Feb 2026) forced a re-glue and re-machine and killed a week of slots in our tightest month (PP-03). This doc collects our mold design rules and adds the one check that would have caught it: someone who didn't design the mold reviews it before any board gets glued or cut.

2. Scope

Female/male molds machined from polyurethane tooling board (30–60 lb/ft^3) for infusion and wet layup. Appendix A covers 3D-printed compression (forged CF) molds. Hotwire foam molds: out of scope (own project doc until standardized).

3. References

Ref	Document	Where
R1	Ure-Bond 90 technical bulletin + SDS	`03 Datasheets/URE_BOND_90_TB.pdf`, `1386A_2-1386B_2.pdf`
R2	*SN4 Mold Manufacturing Problems and Solutions* (2025-10-12)	`SN5 Composites/Documentation/SN4 Mold Manufacturing Problems and Solutions.docx`
R3	*Mold Design Documentation* (wing mold CAD process + videos)	`SN5 Composites/Documentation/Mold Design/Mold Design Documentation.docx`
R4	PP-03 RCA	FEB-COMP-PP-001
R5	CS-005 (machining), CS-004 (sealing)	this series
R6	Rohacell IG-F TDS (pressure/temperature limits for foam cores in molds)	`03 Datasheets/ROHACELL_IG_F_2022_April_EN_243`

Why Ure-Bond 90: SN4's board-stacking method (vacuum-bag with laminating resin) was slow and resin-hungry (R2). Ure-Bond 90 (R1) is a purpose-made rigid urethane adhesive: strong bond between boards, ~4 h clamp time, no bagging. Dowels + spot resin remain acceptable for very large stacks where adhesive cost bites.

4. Definitions

- **Flange**: flat margin around the part cavity for tacky tape, infusion lines, and trimming reference.
- **Scribe line**: machined V-groove marking the part trim boundary and datums.
- **Stack (mold stock)**: tooling boards glued into a block thick enough for the mold.
- **Max-Z**: the machine's maximum cut depth; molds taller than max-Z must be split into sections with alignment datums.

5. Equipment & Materials

Item	Spec / product	Source	Notes
Tooling board	PU, 30 lb/ft³ general / 60 lb/ft³ fine-finish	Coastal Enterprises (sponsor)	60 lb seals better (CS-004)
Board adhesive	Ure-Bond 90	Smooth-On	R1; clamp 4 h
Dowels	Hardwood/aluminum, ~10 mm	McMaster	Large stacks; alignment
Clamps	Bar clamps, even spread	RFS	
CAD/CAM	Fusion 360 (team standard)	—	CAM review per CS-005

6. Safety

- **Tooling-board dust** (gluing prep, any hand-shaping): abrasive and irritating, dust mask minimum, respirator + extraction for power sanding (CS-009 PPE table).
- **Ure-Bond 90** (R1 SDS `03 Datasheets/1386A_2-1386B_2.pdf`): two-part urethane, nitrile gloves, ventilated area, no skin contact; cure heat is mild but clamped assemblies stay put for 4 h.
- Heavy stacks: two people to move anything over ~half a sheet; watch pinch points at clamping.

7. Procedure

7.1 Design rules (CAD)

1. Build the mold body from the part surface per the R3 process (section curves → loft → mirror); keep the machining time target ~3 h per mold section where practical.
2. **Flanges:** 75 mm flat margin beyond the part edge on all sides; tacky tape + spiral tubing + trim allowance need it (SN4 lesson: insufficient side stock, R2).
3. **Scribe lines:** machine part-boundary scribes parallel to ribs/features plus two datum scribes; they must survive sealing (CS-004 §7.3.5).
4. **No overhangs.** 3-axis machining only; check every face has positive draft from the machining direction. Target 2° draft on deep walls.
5. **Foam support:** no unsupported sections thinner than 15 mm; tooling board snaps (R3). Fillet internal corners the finishing-tool radius.
6. **Max-Z check:** if mold height exceeds machine max-Z, split into sections in CAD with dowel/datum features designed in, not improvised at the machine (R2).
7. **Attachment/mounting points:** where the part needs marked hole locations (seat lesson, Oct 2025 meeting notes), design small conical divots into the mold surface.

7.2 Design review: the blocker (PP-03)

1. Before any stock is glued or any slot booked, a reviewer who is NOT the mold's designer walks this checklist against the CAD (10 minutes):

#	Check
1	Part surface matches current part CAD revision (open both; overlay)
2	Flanges 75 mm all around
3	Scribe + datum lines present
4	No overhangs / negative draft from machining direction
5	No sections <15 mm; internal fillets tool radius
6	Height machine max-Z (value in CS-005 §5), or splits + alignment features designed

#	Check
7	Stock size exists (check inventory) and stack plan drawn
8	Machining time estimate written on the WO
9	Stack frozen per CS-002 §7.2 (mold offsets depend on it)

2. Reviewer initials the WO "Mold design review" step. **No initials, no glue, no booking.**

[PHOTO: reviewer overlaying mold CAD on part CAD, checklist on screen]

7.3 Stock gluing

1. Dry-fit boards; mark orientation. Faces must be dust-free (compressed air + IPA wipe).
2. Ure-Bond 90 per R1: mix per TB, spread thin/even on one face, mate, clamp evenly, **4 h clamp minimum**; full cure per TB before machining. For stacks >3 boards or very large molds: dowels + spot resin alternative (R2).
3. **Mark the stack immediately**, in oil-based paint marker, 15 mm characters, on **two adjacent side faces**: part name, WO, and the up-direction (CS-001 §7.4). Mark faces the toolpath will remove less than 6 mm from, or mark the bottom.

   A Sharpie does not survive here. Between glue-up and the CNC slot a glued block sits on a shelf beside three other glued blocks, gets clamped, weighted and IPA-wiped, and the shop's own isopropyl dissolves a Sharpie mark. This is the highest-confusion and lowest-cost-to-fix moment in the whole mold process.

7.4 Machine, clean, seal

1. Machine per CS-005 (zeroing rules live there).
2. Clean and seal per CS-004. Update the WO mold record (layers, density, sealing, location).

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Design review complete	Checklist initialed before gluing	WO blocker step	WO
Glue lines	No visible gaps; no board shift	Visual after clamping	WO step
Machined surface	No steps at section joints > 0.5 mm; scribes crisp	Visual + fingernail	WO qualityChecks
Machining time actual vs estimate	Recorded (feeds future planning, PP-06)	Machine log	WO

9. Records

WO mold block: stack plan, adhesive + clamp time, review checklist initials, machining estimate vs actual, photos at glue-up and after machining.

Appendix A: 3D-printed compression (forged CF) molds

From the SN5 *Forged/Compression Molded Carbon Fiber* study: PLA/PETG molds (40–50% infill, 4–5 walls, 0.1 mm layers) are free at Jacobs but **deform/crack at ~50–100 psi and soften at 60–100 °C**, with no

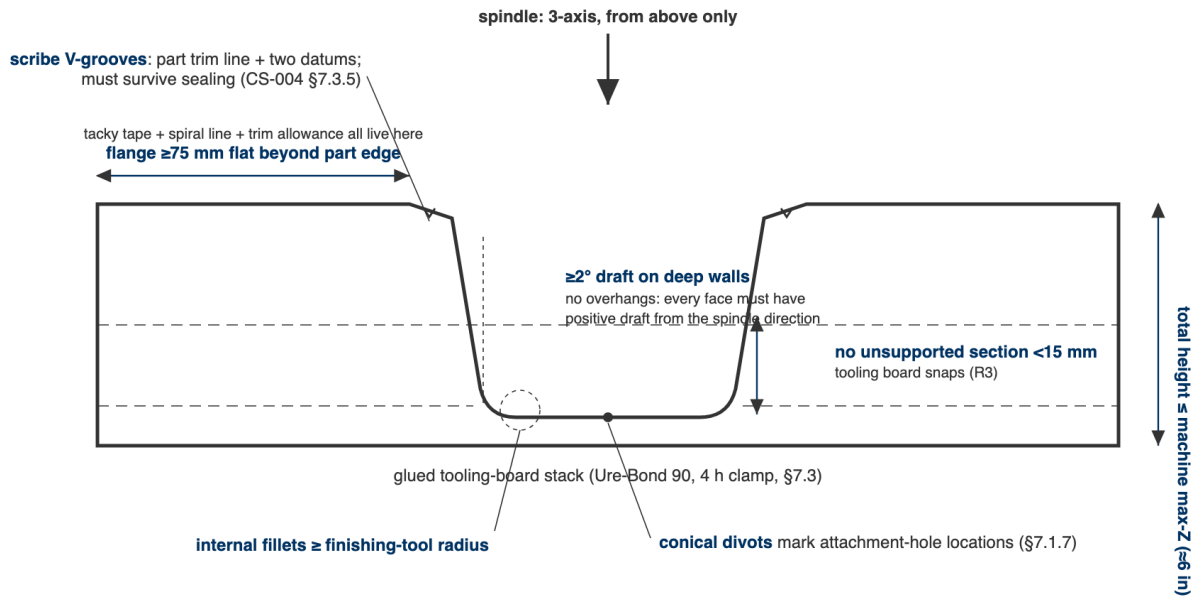


Figure 1: The §7.1 design rules dimensioned on a mold section: flange width, draft, minimum section, fillets, scribes, and the max-Z ceiling, all from the one spindle direction that exists.

hydraulic press and no heated cure. Design: piston-into-cavity, stroke 2× part depth, 0.1 mm sliding-face offset, chamfered split points for wedging, vent grooves against hydrolock. Clamp math: four 600 lbf trigger clamps 2,400 lbf spread over the part area → ~150 psi at 4×4 in (16 in²) but only ~37 psi at 8×8 in (64 in²). Read the two limits together: parts must be small (16–25 in²) for clamps to even *reach* the 50–100 psi consolidation target, and at exactly those pressures a printed mold is at its cracking threshold. So for small forged parts, always distribute the load with through-bolts + aluminum backup plates rather than bare trigger clamps (whose pads also relax over time); for anything larger, clamps cannot reach target pressure at all, so redesign, or accept low-pressure quality. First-part failure mode to avoid: resin starvation (see CS-008 mix math).